PACLAND · Climate, in plain words

What we checked, what we found, and what to do next. One page, no jargon.

1 · The question

Nobody can check a prediction about the year 2100. So the useful question is smaller: can we check whether a climate model deserves to be believed at all?

2 · What we found

- Heat leaks, in most models. Of the models with data to check it, 15 of 19 leak heat. One leaks almost twice as much heat as the warming it is trying to predict. Picture a bucket with a hole: the level you predict is wrong by the size of the hole.
- The simplest guess is hard to beat. Continue the warming we have already measured in a straight line — a ruler. On real data most models lose to the ruler, because a ruler cannot be wrong in complicated ways.
- The leak is in the join, not the ocean. Salt is conserved far better than heat, and salt only moves in water while heat crosses between air and sea. So the mistake is where the model connects the two.

3 · The best we have right now

The model GISS-E2-1-G passes all three checks we can run — how warm it says the world gets, how strongly it traps heat, and whether it leaks. Three out of three, and its certificate says three things checked. On the simplest comparison, the ruler (the warming so far, continued) beats guessing that nothing changes, every time we tried. Whether the ruler or one of the better models wins depends on which years you start from, so we claim the first and not the second.

4 · How we refine it

- Fix the join first. Heat and mass leak and salt does not, so the defect sits where the model couples atmosphere and ocean — a concrete bug in a known place, not a mystery.
- Measure what is missing. Two of the five conservation laws have no measurement in our data yet; closing them makes the checklist five checks instead of three.
- Score against the ruler before anyone uses a model, and never build on a model with no leak data at all — 36 of 53 have none.

5 · What this means for the world

Better projections sooner, by repairing a known defect instead of buying more computers: the missing ingredient is measurement, not number-crunching. And it gives a reader a sentence they can trust in a headline — this projection comes from a model that passes the tests, or it does not.

6 · What this cannot do

It has not checked a single model against the actual future. Nobody has. Every projection is an extrapolation; the difference is whether the model first passed the tests that let you trust the extrapolation.